

Multi-Disciplinary Propulsion Simulations at Engineering Level by means of the European Space Propulsion System Simulation ESPSS

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ABSTRACT

A generic propulsion system simulation tool ESPSS (European Space Propulsion System Simulation) is discussed and applied on various practical systems and engine cycles. The tool is offering the European propulsion community a common platform for easy exchanges of models and numerical representations of propulsion components. The multi-disciplinary EcosimPro Libraries provided with the ESPSS software are described along with the components. The most important aspects of the physical modelling related with the libraries' components are included. The ESPSS consists of multiple libraries to represent a functional propulsion system, e.g. fluid properties, pipe networking including multi-phase fluid flow, two-phase two fluids tanks, non-adiabatic combustion chambers, chemistry, turbomachinery, etc. Four application examples are presented below, one simulating a shock tube case, another related to the evaluation of heat fluxes in combustion chambers, a third one with the simulation of an expander cycle, and a fourth one showing the simulation of a gas generator cycle.

1.0 INTRODUCTION

1.1 Motivation and Goals

A system level prediction for a space propulsion configuration is strongly dependent on proper modelling of each individual propulsion component starting from the pressurant and propellants tanks up to the system's exit e.g. orifices, nozzles... For design, quick turnaround evaluations at operational level, assessment of anomalies... a numerical tool at engineering level is the most appropriate methodology. These tools are classically based upon 0D or 1D model representations of the different propulsion components. Whereas in the past these models were based upon correlations or analytically closed formulations (e.g. pressure drop over a feed-line, bends, pump characteristics...) to predict steady behaviour, they have now gradually evolved towards the solution of the flow properties on the basis of transport equations. These one-dimensional CFD-based (computational fluid dynamics) simulations actually allow a proper coupling between mass, momentum and energy equations including possible extension towards transient phenomena. Where originally the numerical techniques used were based upon Methods of Characteristics (MoC as is still the case for e.g. Flowmaster...), the simulation tools have now moved forward towards pressure or density based one-dimensional discretization techniques. Any research or commercial three-dimensional CFD-package is based upon the same numerical techniques. Its generic nature provides the developer the opportunity to enlarge the tool rather easily with additional transport equation to tackle e.g. multi-phase and multi-components flows. The reduction towards a one-dimensional system representation provides the user the potential of performing multiple simulation runs within a reasonable time. Simultaneously, this provides the opportunity to couple it with other disciplines such as conjugate heat transfer, chemistry.... which often needs to be taken into account for an accurate numerical prediction.

Over the last decades, the aerospace industries and related research institutions have developed dedicated in-house simulation tools to correctly predict the behaviour of components, sub-systems up to complete

systems. This resulted into a wide variety of numerical tools which differ from each other by particularities in modelling various specific components. This had led to individual simulation strengths linked to in-house competences and experience built up over the years for particular components or situations. Having direct access to the source code of these tools is undeniable a prerequisite for the aerospace community. This guarantees quick implementation of particular refinements often needed for new propulsion components, encountered difficulties, anomalies or future extensions. However, the intrinsic danger of this vast set of dedicated propulsion components simulation tools pops up when all of these components need to be physically linked to each other in case an overall system level study is required. Unfortunately, each of these dedicated tools, having a proper database, a dedicated numerical scheme etc... are generally not tuned to work together. This is particularly the case when the various subsystems are modelled and validated by different companies. As a result any newly devised propulsion system involving components from different providers requires the development of custom-made interfaces and communication protocols. This is clearly a cumbersome and expensive exercise. Harmonization of the different tools or the creation of specific standards could clearly help the propulsion community.

Based upon the above mentioned observations, the idea grew at ESA-ESTEC (the European Space Technology and Research Centre) to create a common, generic simulation platform for the European Space Propulsion community. Originally started as a simulation tool for spacecraft propulsion system with the PropSat library as prototype, it quickly attracted also the attention and interest from major European aerospace industries such as Astrium, Avio, Thales... for satellite and later on for launcher application. This gradually evolved towards the development of the European Space Propulsion System Simulation ESPSS. The general idea was to:

- offer a *common simulation platform* to the European Space Propulsion Community
- provide a common propulsion database wrt to propellant fluid properties, chemistry...
- smoothen the exchange of component or sub-systems models without difficulties with respect to interfaces, databases, ... for joint European and national projects
- offer a tool able to be used by different levels of users i.e.: library developer, system modeller, operation engineer and external operation engineer (black box)
- allow connection to other advanced tools, i.e. zooming capabilities such as CFD, thermal analysis...
- provide an *open source* code
- to provide as a basis a *generic representation* of propulsion components available from open literature

The last two points avoid any infringement of in-house competence in terms of modelling. Companies are not forced to provide their modelling particularities but have the option to implement those themselves or to exchange them as black boxes. The on-going development of ESPSS ensures continuous validation both for generic test cases and complex applications. This philosophy ascertains general use and maximum flexibility.

The present lecture note is actually a compilation of various articles published related to the development, validation and/or application of the ESPSS library from the start till September 2012 [1] [2] [3] [4] [5] [6] [7] [8] [10].

1.2 Background

The highly iterative process that characterises the developing of a space transport system could benefit from an inexpensive tool capable of providing useful qualitative and quantitative indications as early as the preliminary design. When compared to test campaigns, numerical simulations are inexpensive. However, test campaigns are fundamental in the design process and are also required by international regulations. Therefore simulation tools are not intended to substitute tests, but rather to reduce the number of tests needed, by guiding the design process in the concept phase and preliminary design [10].

In order to have a better understanding of how much test campaigns and transient problems could have an

impact on the development phase of an engine, a description of the main failures for the Japanese LE-7 engine and for the American SSME engine are here described.

The Space Shuttle Main Engine (SSME) was used in the Space Shuttle. For this very complex engine 5 years of analysis were necessary to model the transient behaviour of the propellants and of the hardware during start-up and shut-down. The SSME engine was sensitive to small changes to propellant conditions and valve tuning was critical, a 2% error in the valve position or a 0.1 s timing error could lead to significant damage to the engine. A step-by-step approach was necessary to explore the start-up sequence with small time increments: that required 19 tests, 23 weeks, 8 turbopump replacements to reach 2 s into 5 s of start-up. Additional 18 tests, 12 weeks and 5 turbo-pump replacements were necessary to touch the minimum power level [42].

The LE-7 is a high performance LOX/LH₂ engine employed in the H-II rocket. During H-II Flight 8 in 1999 a failure occurred in the LE-7 engine. The failure was determined by mechanical vibration problems into the fuel pump that caused high cycle fatigue and so the premature engine failure. During development tests of the LE-7 other issues can be addressed to phenomena occurring during transient phases:

- High sensitivity to heat transfer from hardware to fluids and to ignition timing reproducibility of both combustion devices
- Functional instability to inlet pump vibration of pressure
- Problem of rotating cavitation on oxidiser and fuel pumps
- Over power during start-up caused damages to hardware
- Explosion at Main Oxidiser Valve opening

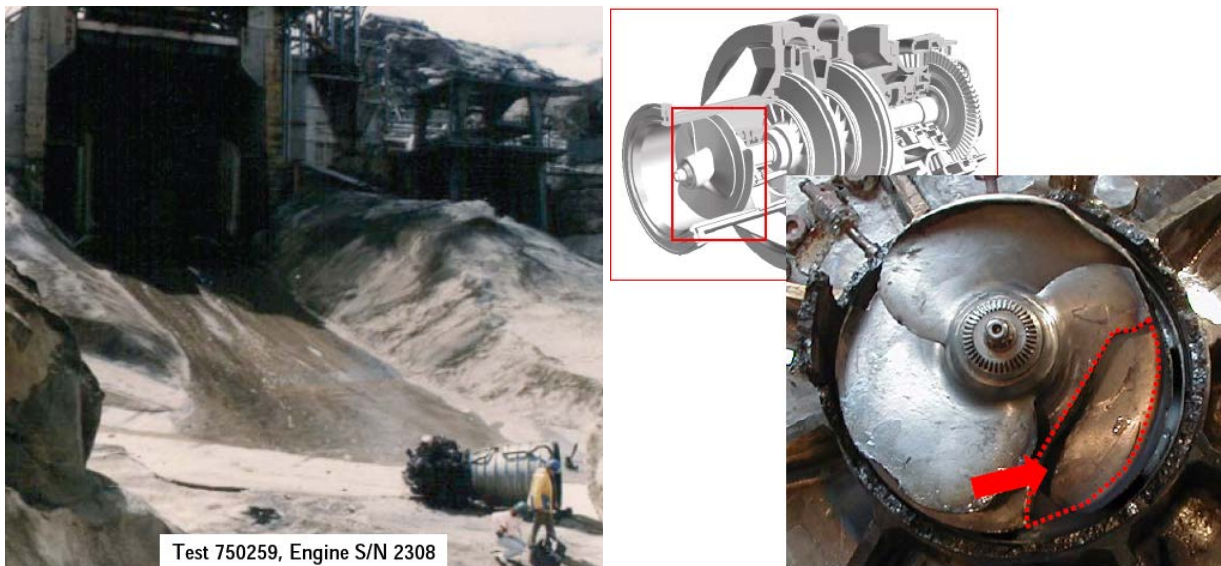


Figure 1: SSME and LE-7 failures

Clearly, liquid propulsion systems are very complex systems. Modelling the whole propulsion system is possible by identifying more “simple” subsystems (or components) and connecting them in a congruent way. Each component interacts with the others by means of inputs and outputs, but its dynamical behaviour, that is, the way the inputs are treated to give the outputs, is characteristic of the component. This leads to the paradigm of Object Oriented Modelling (OOM): the whole system is modelled by connecting the single subsystems. The main advantage of this approach is an enhanced of manageability and of organization of the model towards implementation in a numerical solver [11].

One of the latest Object Oriented Programming (OOP) languages is the EcosimPro Language (EL) [1] built in Ecosim-Pro, developed by Empresarios Agrupados International. EcosimPro is an analysis tool that combines OOP and OOM methodology to model various kinds of physical systems which can be represented by differential-algebraic equations (DAE) or ordinary differential equations (ODE). Within this framework, objects can be modelled independently from each other and then be interconnected by ports to model more complex dynamic systems. Ports represent the interface between the components, where the inputs and outputs are transmitted. OOP makes it possible to create libraries in which both abstract and child components related to a particular system or a physical field are grouped. The improvement with respect to other programming languages is that EL makes it possible to model both continuous and discrete events. Continuous statements are used to express sets of differential-algebraic equations, where the order in which they are written is not important, as they will be sorted later by the internal algorithms. They form the core where the continuous physical models are defined; discrete statements are used to express events. These events are controlled by conditional statements which indicate when an event occurs.

The DASSL numerical method and the fourth order Runge-Kutta method are the two built-in numerical solvers present in EcosimPro. The former enables to solve the DAEs, whilst the latter solves the ODE. ESPSS (European Space Propulsion System Simulation) is the full set of libraries written to model a functional space propulsion system. Among the libraries collected in ESPSS, there are fluid properties, pipe networking, tanks, combustion chambers etc.[1],[12] Comparing EcosimPro and its OOM paradigm with other numerical simulation tools such as Computational Fluid Dynamics (CFD), results in a computational effective modelling but at a cost of a lower spatial resolution. This is acceptable if the task of the simulation is a system study. Nonetheless, a system approach is necessary to take all the interactions into account between the different components of a space engine. Though, hypothetically possible with CFD, it is practically not justified at this level. However, if particular components need to be represented by a high-fidelity code such as CFD but within the overall system analysis, each of these components can be replaced by a high-fidelity model. This is referred to as ‘zooming’ [7].

2.0 OVERVIEW OF ESPSS LIBRARIES

The following libraries have been developed as part of ESPSS: ***fluid properties***, ***1-D fluid flow***, ***tanks***, ***combustion chambers*** and ***turbomachinery*** libraries. An overview of these propulsion libraries is presented here below. More details can be found in [1], [4], [12].

2.1 Fluid Properties

The fluid properties library is in charge of the calculation of fluid properties. Functions available on this library are mainly used by the 1-D fluid flow library for the simulation of fluid systems. Three main classes of fluids are available:

- Ideal gases, with temperature dependent thermodynamic and transport properties
- Simplified liquids, with temperature dependent properties
- Real fluids, with tabulated properties depending on both temperature and pressure

where the last class covers liquid, superheated, supercritical and two-phase fluids. Two-phase, two-fluid mixtures of a real fluid in any thermodynamic state with a non-condensable (ideal) gas are allowed. The homogeneous equilibrium model is used to calculate the properties (quality, void fraction, etc...) of a real fluid in two phase conditions, with or without a non-condensable gas mixture. User-defined fluids are available for all fluid classes, which can be defined in external data files. The fluid properties library does not contain any component. Instead, it provides a large collection of functions returning the value of a fluid property (or the complete thermodynamic state) by introducing relevant parameters. In particular, the complete fluid state is defined by any combination of two thermodynamic state variables (temperature,

pressure, density, internal energy, enthalpy, entropy, etc...).

Though the users can easily add any fluid of interest, some basic propulsion related fluids are already implemented based upon publicly available data, i.e.:

- Storable propellants: MMH, NTO, MON-1, MON-2
- Cryogenic propellants: H_2 , O_2 , CH_4 ...
- Pressurants: He, N_2
- General: air,...

2.2 1-D Fluid Flow

The 1-D fluid flow library allows transient simulation of two-fluid, two-phase flow systems. It is linked to the fluid properties library described in the previous section, which provides the necessary functions for assessing fluid conditions. Flow inversion, inertia and high speed phenomena are considered in pipes, volumes and junctions. Pipes are also incorporating an area-varying non-uniform mesh 1-D spatial discretisation into n volumes.

Concentrated (valves) and distributed (pipes) load losses are modelled, and heat transfer between metallic walls and the fluid can be taken into account. Multiple thermo-hydraulic correlations and initialization options are included. Other special components such as check valves, pressure regulators, heat-exchangers and T-junctions are available. 1-D pipe flows can be simulated using some of the most up-to-date, robust and accurate CFD techniques: upwind (Roe) or centred schemes.

Hydraulic or pneumatic complex systems where heat transfer and control are coupled are easily evaluated with the 1-D fluid flow library. Bubble formation due to the cavitation phenomena or to the presence of a non-condensable gas in a liquid is calculated in pipes or other components. Besides, the 1-D fluid flow library permits to analyse in great detail transient aspects due to inertia (water-hammer) and bubble collapse.

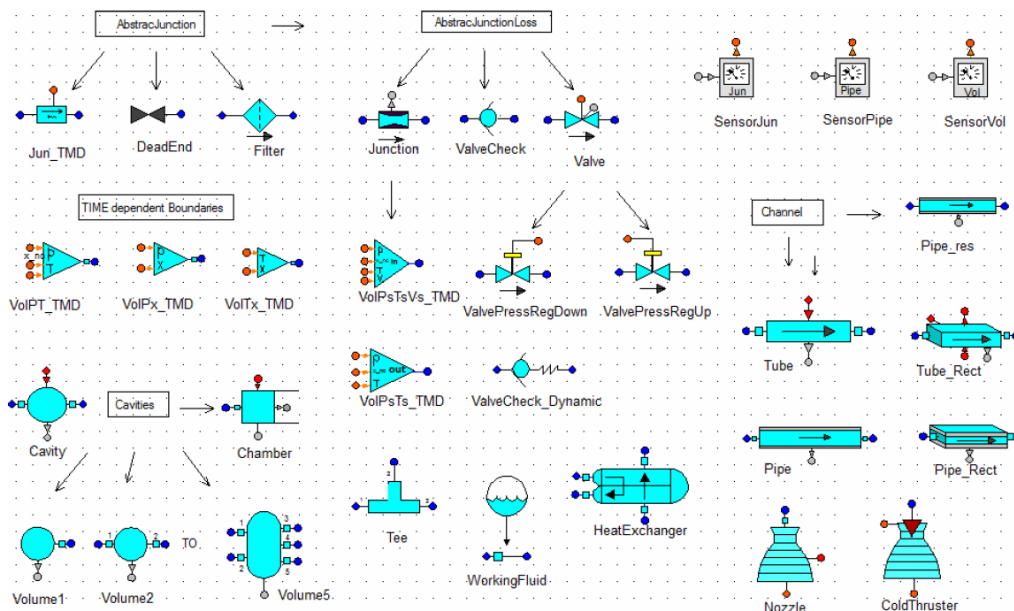


Figure 2: 1-D fluid flow symbol palette

Component overview. Figure 2 shows the main library components. In an EcosimPro fluid network, every component is either a resistive component or a capacitive component. A resistive component receives the

state variables (pressure, density, velocity, chemical composition and enthalpy) as input in inlet and outlet and gives back the flow variables (volumetric, mass and enthalpy flows) as output. A capacitive component receives the flow variables as input and gives back the state variables at output. To build a fluid network, the user must connect resistive components to capacitive ones, alternatively. So, from a computational point of view, components are divided into two classes:

- C (capacitive) elements, integrating the mass and the energy conservation equations. Thermodynamic functions are used to calculate the complete thermodynamic state
- M (momentum) elements, calculating explicitly (inertia terms) the mass flows between C elements. Reverse flow is allowed

This computational scheme prevents the appearance of non-linear algebraic loops and high index DAE (Differential Algebraic Equations) in the mathematical model of the pipe network.

Pipe component. As an example, a detailed description of the pipe modelling is given. This component simulates an area-varying non-uniform mesh 1-D pipe that exchanges heat with a 1-D thermal port. The general case of a mixture of two fluid components, for which the first one can be either one phase or two-phase, and the second one is non-condensable gas, is simulated through the solution of the following 4 conservation equations:

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial f(\mathbf{u})}{\partial x} = \mathbf{S}(\mathbf{u})$$

$$\mathbf{u} = A \begin{pmatrix} \rho \\ \rho x^{nc} \\ \rho v \\ \rho U \end{pmatrix}, \quad f(\mathbf{u}) = A \begin{pmatrix} \rho v \\ \rho x^{nc} v \\ \rho v^2 + p \\ \rho v H \end{pmatrix},$$

$$\mathbf{S}(\mathbf{u}) = \begin{pmatrix} -\rho A K_w \frac{\partial p}{\partial t} \\ -\rho x^{nc} A K_w \frac{\partial p}{\partial t} \\ -\frac{1}{2} \frac{\Delta \xi}{\Delta x} \rho v |v| A - \rho g A + p \frac{dA}{dx} \\ \frac{\Delta Q}{\Delta x} \end{pmatrix}$$

This set of equations is closed by a thermodynamic equation of state (EoS), which is described in the fluid properties library, and hereafter written under general form:

$$p = p(\rho, u)$$

The choice of density ρ and internal energy u as independent thermodynamic variables is the most efficient one regarding CPU-time when the EoS is left under arbitrary form. The different source terms are the following:

- In the first equation governing the mixture mass conservation, a source term responsible for the wall compressibility effect of the mixture, determinant in water hammer simulations, is included. This wall compressibility K_w takes into account the type of anchoring at both ends and results in different functions with the following form:

$$K_w = f(D, t, E_Y, \nu)$$

- In the second equation governing the non-condensable mass conservation (if any), a similar source term is included;
- In the third equation governing the mixture momentum conservation, a source term represents the equivalent distributed friction: $\Delta \xi$, proportional to Δx , is the pressure drop coefficient derived by the following empirical correlation:

$$\Delta\xi_i = \frac{k_{add}}{n} + \sum_j \frac{\xi_{bend,i,j}}{n} + \frac{\Delta x_i}{D_i} f_{fr,i}$$

where k_{add} is an input data representing some concentrated load losses to be distributed along the pipe. $\xi_{bend,i,j}$ is the bend pressure drop coefficient for node i and bend j [13]:

$$\xi_{bend,i,j} = f(\alpha_j, Re_{c,j}, D_i, \varepsilon)$$

$f_{fr,i}$ is the friction factor, calculated by means of a correlation valid for laminar, turbulent and transition flow [13]:

$$f_{fr,i} = f(Re_i, D_i, \varepsilon)$$

- In the same equation, another term accounts for gravity, and the last one is responsible for the area variation;
- In the last equation governing the mixture energy conservation is included a source term ΔQ (proportional to Δx , see below) taking into account the heat exchange with the wall through a heat port is included:

$$\Delta Q_i = h_{f,i} \Delta x_i P_{in} (T_{out} - T_i)$$

the convective film coefficient h_f is calculated with different empirical correlations [14],[15] depending on the fluid state and wall temperature (single phase, condensation, superheated condensation, boiling, subcooled boiling...). For single phase, turbulent flow, the well-known correlation is used [16]:

$$Nu = 0.023 Re^{0.8} Pr^{0.4}$$

The implementation of this set of equations must take into account various parameters, i.e. geometry (flow area and dx can vary along the pipe), numerical scheme, boundary conditions, and flow thermodynamics (state law, composition, etc).

Numerical schemes. The tube component is discretised by either a centred or an upwind numerical scheme. Figure 3 describes the pipe discretisation. The inner fluxes are computed using one or the other of these schemes, and the first and last junctions (1 and $n + 1$) ones are given by the fluid ports, as they are calculated at resistive type components using momentum equation with sonic flow limitation.

Note that the first and last half-nodal inertia are included in the junction component equations. Using the centred scheme, a staggered mesh approach is applied, for which the state variables (pressure, density, velocity, chemical composition and enthalpy) are associated with the n nodes, and the flow variables (volumetric, mass and enthalpy flows) are calculated at the internal junctions (each junction has associated two half volume inertias).

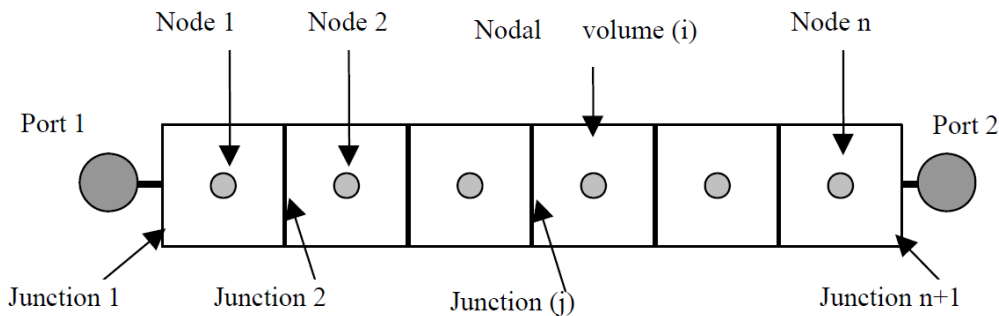


Figure 3: Pipe discretisation

With this scheme, the various fluxes to be computed at the inner junctions are simply the flow variables, except for the mixture momentum flux that is associated to the n nodes:

$$\begin{aligned} f_{\text{mass},j} &= \dot{m}_j \\ f_{\text{nc},j} &= \dot{m}_j^{nc} \\ f_{\text{mom},i} &= (p_i + qn_i + \rho_i v_i^2) A_i \\ f_{\text{ene},j} &= \dot{m} H_j \end{aligned}$$

The momentum flux term includes an artificial dissipation. As an alternative to the centred scheme, an adequate upwind scheme has been developed to improve the discontinuities resolution of these transient two-phase flows in quasi-1D pipe networks. Using that scheme, a collocated mesh approach is applied, for which all variables are discretised on the n nodes, even the mass flow.

All the fluxes $f(\mathbf{u})$ are discretised at the junctions and include a central part and an upwind part, following the initial Roe scheme [17]; the scheme has been then adapted to handle arbitrary equation of state: the pressure derivatives of the Jacobian were written as function of various tabulated fluid properties (κ , β , c_v , etc.) and linearised using a clever two step projection-correction method. The projection step computes an approximation of the linearised pressure derivatives by some numerical integration (midpoint, trapezoidal, simpson), and the correction step checks if these approximations verify some coherence criteria, and correct them accordingly. Moreover, these linearised pressure derivatives and the Roe averages of primitive variables are computed taking into account the possible occurrence of a phase transition between the left and the right state, by using a piecewise linear integration path following the saturation line, if any.

For steady low Mach number computations, a Turkel preconditioning can be applied, whereas for unsteady low Mach number computations, a simple pressure correction can be performed. The MUSCL high resolution method can be applied to all the variables to reach a third order precision in the smooth parts of the flow, and to limit them to first order near discontinuities, using various limiters.

Finally, if needed, some source terms can also be upwinded following the same methodology as for the flux terms, in order to provide a full well-balanced scheme.

2.3 Tanks

The **tanks** library enables the transient simulation of rocket engine and spacecraft tanks. As in the **1-D fluid flow** library, gas, liquid and two-phase flow regimes can be modelled inside a Tank for ideal or real fluids. The homogeneous equilibrium model is used to calculate a real fluid in two phase conditions, with or without a non-condensable gas mixture.

Most typical wall shapes have been included as separated components to model heat conductivity in walls and insulations. Multiple correlations are available [18], [19], [20] to calculate the heat transfer between the tank wall and the fluid cavities, including boiling conditions. Single tank models working with a pressurant gas and a liquid are available, including liquid level calculation.

Rocket engine pressurization systems where the feeding tanks play an important role are easily evaluated using this library. Tanks components can be connected with 1-D fluid flow components with the aim of simulating a complete rocket engine cycle.

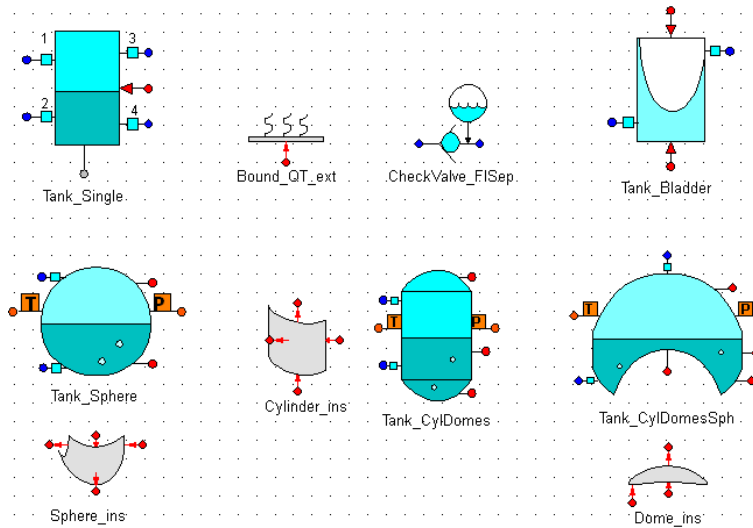


Figure 4: Tanks symbol palette

Component overview. Figure 4 shows the main library components. The bladder tank component includes the bladder motion calculation between two separated fluid volumes. The 1-D tank components consider a 1-D moving grid discretisation for the liquid side and another for the gas cavity. The walls have fixed 2-D spatial discretisation. Heat exchange at the gas/liquid/wall interfaces is simulated. Heat and mass exchange is also considered at the liquid/gas interface.

2.4 Combustion Chambers

The *combustion chambers* library enables the simulation of rocket engines and thrust chamber elements. Combustion gas mixture properties (transport and heat capacity) are calculated from adequate coefficients [21] from each chemical species present as combustion product. Minimisation of Gibbs free energy is applied to find equilibrium molar fractions at of a mixture of reactants.

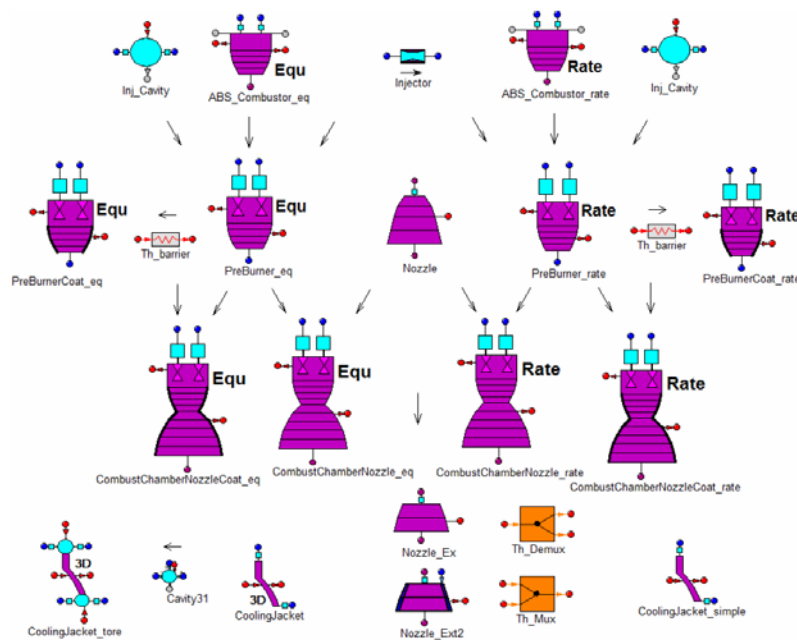


Figure 5: Combustion chambers symbol palette

Component overview. Figure 5 shows an overview of all combustion chambers components. 1-D combustion chambers, 1-D nozzles, and regenerative circuit components are modelled. The main components are:

- A 1-D non-adiabatic combustor component, where equilibrium combustion products are calculated at each node using the Gibbs approach. This element is described in detail hereunder.
- A more advanced combustor component taking into account non-equilibrium effects, by using vaporisation and global reaction time constants.
- Several combustion chamber components built from a combustor, two injectors (junctions) and two cavity (volume) components, as shown in Figure 5
- two models of regenerative circuits or liquid cooling jackets, with 2-D or 3-D walls, respectively.
- Ideal or non-adiabatic 1-D exhaust nozzles.

As an example, the combustion chamber component will be described in more detail.

Combustion chamber component. This component represents a non-adiabatic 1-D combustion process inside a chamber for liquid or gas propellants. The equilibrium combustion gases are calculated for each chamber volume (node) using the minimum Gibbs energy method as a function of the propellant's mixture molar fractions, inlet conditions and chamber pressure.

Transient chamber conditions (pressures, temperatures, mass flows and heat exchanged with the walls) are derived from 1-D transient conservation equations. A mixture equation between the injected propellants and the combustion gases is applied. From the definition of the mixture ratio MR and derivation, the following dynamic equation gives the MR evolution:

$$\dot{m}_{ox} = MR\dot{m}_{red} + \frac{d}{dt}(MR) \frac{\rho V_{ch}}{1 + MR}$$

Combustion takes place when mixture ratio is within the allowed limits, the ignition flag is active and a minimum time (ignition delay) τ has elapsed. More sophisticated models can be programmed to simulate more accurate ignition processes.

Mass, energy and momentum equations are basically the same as in the pipe component with variable cross area. The centred scheme is used to discretise the chamber, using a staggered mesh approach. Hence, node velocities are calculated using the adjacent junction mass flows and the mean junction densities:

$$v_i = \frac{1}{A_i} \left(\frac{\dot{m}_{jun,i-1}}{\rho_i + \rho_{i-1}} + \frac{\dot{m}_{jun,i+1}}{\rho_i + \rho_{i+1}} \right)$$

For each chamber volume, the combustion products properties (molar fraction, heat and transport properties) are calculated using the minimum Gibbs energy method as a function of the propellant molar fractions, volume pressure and volume specific enthalpy. In fact, the propellants can be formed by any allowed fluid mixture using the **1-D fluid flow** library, including that of a previous combustion.

Once the first volume is calculated, subsequent volumes will consider that the molar fraction of the products of the previous volume will act as the inlet propellant mixture, so the Minimum Gibbs energy method can be applied to any combustor volume. The term ΔQ_i appearing in the energy conservation equation permits the exchange of heat through a thermal port. The walls (that can be represented by thermal components or the Cooling Jacket component) are not included in this component, but instead are taken as a boundary for the heat exchange calculation:

$$\Delta Q_i = h_{f,i} A_{wet,i} (T_{aw,i} - T_{out,i}) + \sigma (A_{wet,i} (T_i^4 - T_{out,i}^4))$$

where both convective and radiative heat are taken into account. Here, T_i is the static temperature, $T_{out,i}$ is the wall temperature and $T_{aw,i}$ is the adiabatic wall temperature defined as:

$$T_{aw,i} = T_i \left(1 + Pr_{ref,i}^{0.33} \frac{\gamma_i - 1}{2} M_i^2 \right)$$

$$Pr_{ref,i} = \left(\frac{c_{p,i} \lambda_i}{\mu_i} \right)_{ref}$$

Reference conditions *ref* are calculated at a temperature halfway between the wall and the free stream static temperature. The mixture composition is assumed to be constant between these two temperatures. The convective film coefficient is calculated using semi-empirical correlations according to Bartz [22]:

$$h_{f,i} = 0.026 \mu_{ref,i}^{0.2} \left(\frac{\lambda}{\mu} \right)_{ref,i}^{0.6} c_{p,ref,i}^{0.4} \frac{\dot{m}_{th}^{0.8}}{A_i^{0.9}} \left(\frac{\pi D_{th}}{4 R_c} \right)^{0.1}$$

2.5 Turbomachinery

Turbomachinery is an EcosimPro library for the simulation of pumps, turbines and compressors.

Pump components are provided with dimensionless turbopump characteristics curves adapted to positive and negative speeds and flow zones. Turbine and compressor components are provided with dimensionless performance maps as a function of the reduced axial speed and pressure ratio. **Turbomachinery** components allow using specific user provided performance maps. As an alternative, if the full turbopump characteristics are unknown, simplified (design) components allow the calculation of generalised performance maps as a function of the nominal performances and other significant design data.

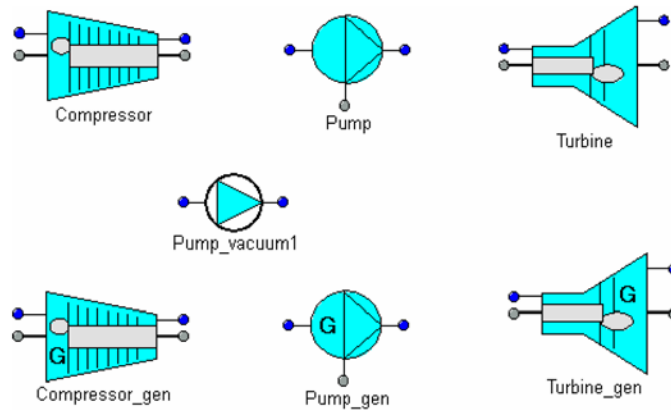


Figure 6: Turbomachinery symbol palette

Component overview. Figure 6 shows the main library components. **Turbomachinery** components are built topologically with a **1-D fluid flow** volume component receiving/giving the head rise and the mechanical work, and a junction simulating the outlet.

- The performance maps (mass flow coefficient and specific torque) are introduced by means of 2-D input data tables using dimensionless variables adapted to the different types of components
- Equations used to calculate head and mechanical work are based on the performance maps, and are dimensionless with respect to the inlet of the turbomachinery volume

As an example, the generic pump component will be described more in detail hereafter.

Generic Pump component. This component simulates a pump for liquids. It is provided with fixed or user-defined dimensionless turbo-pump characteristic curves adapted to positive and negative speeds and flow zones, and valid for several types of pumps. Usually it is rather difficult to find pump curves including the non-normal zones, so this is the reason to include in this component a set of curves covering all zones of the

pump operation for 3 different specific speeds: $N_s = 25$ corresponding to a pure centrifugal pump, $N_s=147$ corresponding to mixed pump, and $N_s=261$ corresponding to an axial pump.

The specific speed is defined as:

$$N_s = \frac{rpm \sqrt{Q/n_s}}{(TDH/n_{st})^{0.75}}$$

where n_s is the number of suctions, n_{st} is the number of stages, Q is the volumetric flow (m³/s) and TDH represents the actual total dynamic head of the pump (m).

The pump curves (head and torque) are introduced by means of fixed 2D data tables named “wh_vs_theta_Ns” and “wt_vs_theta_Ns” defined as functions of the specific speed and a dimensionless variable θ that preserves homologous relationships in all zones of operation. The θ parameter is defined as follows:

$$\theta = \pi + \tan^{-1}(v/n)$$

“v” and “n” are the reduced flow and speed variables:

$$v = Q/Q_R = m_{in}/\rho_{in}/Q_R; \quad Q_R : \text{Nominal volumetric flow}$$

$$n = \frac{30\omega/\pi}{rpm_R}; \quad rpm_R : \text{Nominal speed (rpm)}$$

The dimensionless characteristics (head and torque) are defined as follows:

$$wh = \frac{TDH}{TDH_R(n^2 + v^2)}; \quad wt = \frac{T}{T_R(n^2 + v^2)}$$

T and TDH are the torque and the total dynamic head respectively. Sub index R means “rated” (nominal) conditions (input data). The nominal torque is calculated from the other nominal parameters:

$$T_R = \frac{9,806 \rho_{up} TDH_R Q_R}{\text{eff}_R rpm_R 2\pi / 60}$$

The dimensionless performances will be calculated interpolating in the aforementioned curves for the actual values of the specific speed N_s and θ .

Using the definition of the wh, wt, the actual torque T and the pressure rise TDH (expressed as the total dynamic head in meters) will be calculated.

The mechanical balance allows the calculation of the axial speed dynamically:

$$I_{mech} \cdot \dot{\omega} = T_{shaft,port} - T$$

The mechanical work (Torque*w) received by the pump volume is simulated as an enthalpy flow.

$$\text{Pump.Volume.mh} = \text{Inlet_Pump.mh} + T \cdot \omega$$

where the variable mh is the enthalpy flow. It must be emphasized that this formulation with an internal volume receiving the work allows the simulation of the fluid heating in every situation, even when the pump works without flow (null efficiency).

The current efficiency in off-design conditions can be calculated accordingly with the definition of the nominal torque (isentropic enthalpy rise assumed to be the pump head, $9.806 \cdot TDH = (h_{is} - h_{in})$). Then,

$$\eta = m \cdot 9.806 \cdot TDH / T \omega$$

Finally, the inlet mass flow equation is expressed dynamically in accordance with the pump pressure rise as follows:

$$I \cdot \dot{m}_{in} = P_{out} - (P + 0.5 \rho v^2)_{Pump, vol} + 9.806 \cdot \rho_{in} \cdot dtdh_{cav}(NPSH, Q)$$

3.0 VALIDATION AND APPLICATIONS

3.1 Academic/Advanced components

3.1.1 Shock tube

The implemented Roe scheme and its appended numerical developments have been validated extensively during the development phase. This section presents the benefits of that scheme for a multi species compressible flow. The Karni and Quirk shock/bubble interaction test case [24] has been selected for validation. It consists of a shock-tube filled with air ($\gamma_{\text{air}} = 1.4$, $R_{\text{air}} = 287.1 \text{ J/kg}\cdot\text{K}$), with a right-travelling shock wave. In the pre-shock wave region, a bubble of helium is introduced with a finite length ($\gamma_{\text{He}} = 1.67$, $R_{\text{He}} = 2077.15 \text{ J/kg}\cdot\text{K}$). The test case setup and initial (dimensionless) conditions are shown on figure 7.

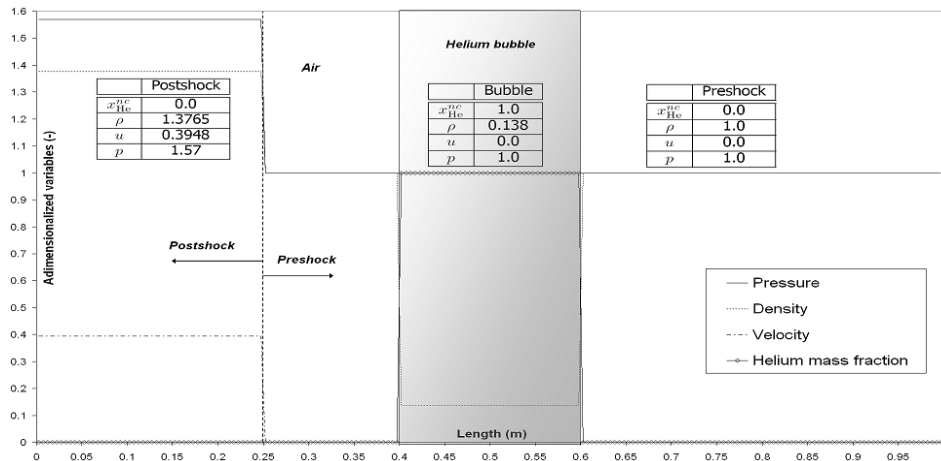


Figure 7: Initial conditions of shock/bubble interaction test case

The shock wave is initially located at $x = 0.25$, and the bubble sits between $x = 0.4$ and $x = 0.6$. Simulations have been run with the centred scheme and the upwind Roe scheme described in a previous section. Figure 8 compares pressure profiles at the dimensionless time $t = 0.35$, using the centred scheme (dotted line) and the upwind Roe scheme (dashed line), together with the exact solution for this test case (continuous thick line). The mesh is uniform and contains 200 cells.

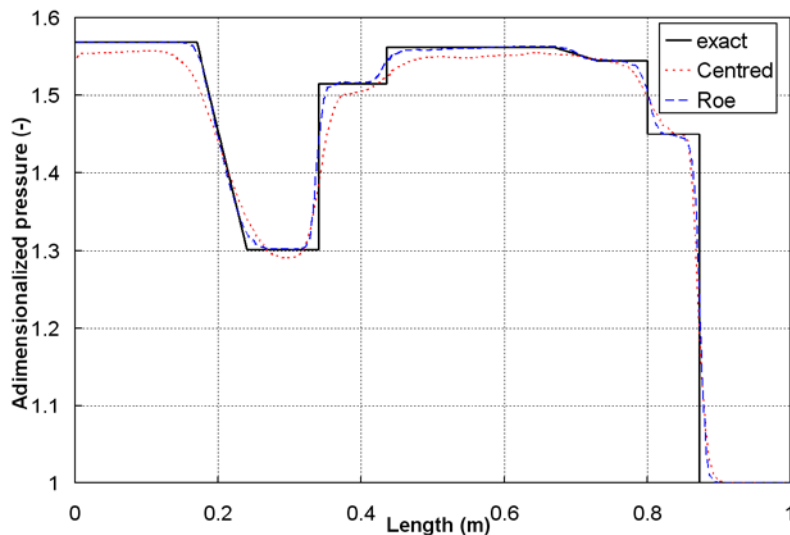


Figure 8: Shock-tube bubble interaction test case: Pressure

It can be seen on the plots that the centred scheme underestimates the pressure value, and is poorly

resolving the discontinuities. On the contrary, the upwind scheme performs well in resolving the various single component (shocks and rarefactions) and multi component (shock/bubble interaction) discontinuities, as well as the constant value regions. Pressure oscillations likely to occur in conventional conservative schemes applied to multi species compressible flows [25] have been avoided thanks to the primitive variable reconstruction.

3.1.2 Hot Gas side heat transfer coefficient models [10]

In the years 1999-2000, in the frame of an ESA GSTP-2 contract, Astrium performed a series of experiments with a water cooled calorimetric combustion chamber [26]. Different correlations based on Bartz ([26],[22]) and Pavli ([26],[27]) have been implemented to simulate the calorimetric combustion chamber tests and their results have been compared with the experimental results from this calorimetric chamber test campaign [28], [29]:

$$St_{Bartz} = 0.026 \left(\frac{\mu_{ref}^{0.2}}{C_{p,ref}^{0.6}} \right) \left(\frac{\lambda_{ref}}{\mu_{ref}} \right)^{0.6} (\dot{m})^{-0.2} A^{0.1} \left(\frac{D_{th}\pi/4}{R_{curv}} \right)^{0.1}$$

$$St_{Pavli} = 0.023 Re^{-0.2} Pr^{-0.6} \left(\frac{T_{aw}}{T_{ref}} \right)^e \left(\frac{x}{x_{th}} \right)^f$$

To improve the behaviour of the original Bartz equation, a temperature correction factor K_T was added, taking into account that the reference temperature is calculated halfway between the wall and the free stream static temperature. Moreover, since the geometric reference parameter in the original Bartz equation was the throat diameter, a further correction factor K_x was added for the consideration of the boundary layer growth in the cylindrical part and in the nozzle [22]; other two correction factors were then added: one taking into account the vaporization phenomenon near the injector plate, the so-called “combustion zone”, and the other has been added to achieve a better agreement between the numerical and the experimental results in the nozzle region; this correction factor K_{acc} is related to the flow acceleration caused by the nozzle contour [10].

The calorimetric chamber is a sub-scale, water cooled thrust chamber with twenty segments. Each segment features an independent water feed system with volume flow measurement. For each segment the heat flux is measured individually. The described calorimetric system has been modelled using the following components [30]:

- 1 combustion chamber with 21 nodes (component to validate)
- 20 regenerative circuits with 5 nodes each
- 20 mass flow regulated water feed lines (with the necessary junctions and boundary conditions)
- mass flow regulated propellant feed lines

The simulation was performed using the couple liquid oxygen/gaseous hydrogen as propellants, at an O/F ratio varying from 5 to 7 and at a total pressure from 35 to 70 bar in the combustor; for each test point the propellant mass flows are chosen in order to get the desired pressure and mixture ratio. In order to get the right pressure drop through the cooling circuit, the roughness of the cooling channels had to be adapted. Values between 3.2 and 25 μm were chosen. This tuning was necessary because of the partially unknown layout of the cooling circuit and its feed lines (pipes, fittings...).

The heat fluxes calculated with every correlation described above are plotted in Figure 9 where “ q_{w_ref} ” is the heat flux at the throat. Figure 9(a) describes the results for the nominal case ($p = 60$ bar, $MR = 6$); in Figure 9 (b) simulation results are compared to experimental data obtained varying the chamber pressure ($p = 35; 60; 70$ bar, $MR = 6$), Figure 9 (c) shows the simulation and experimental comparison for tests with constant chamber pressure but different mixture ratio ($p = 60$ bar, $MR = 5; 7$), while in Figure 9 (d) the hot gas wall temperature trend is shown at different chamber pressures. The correlations are all plotted with lines and experimental data with symbols.

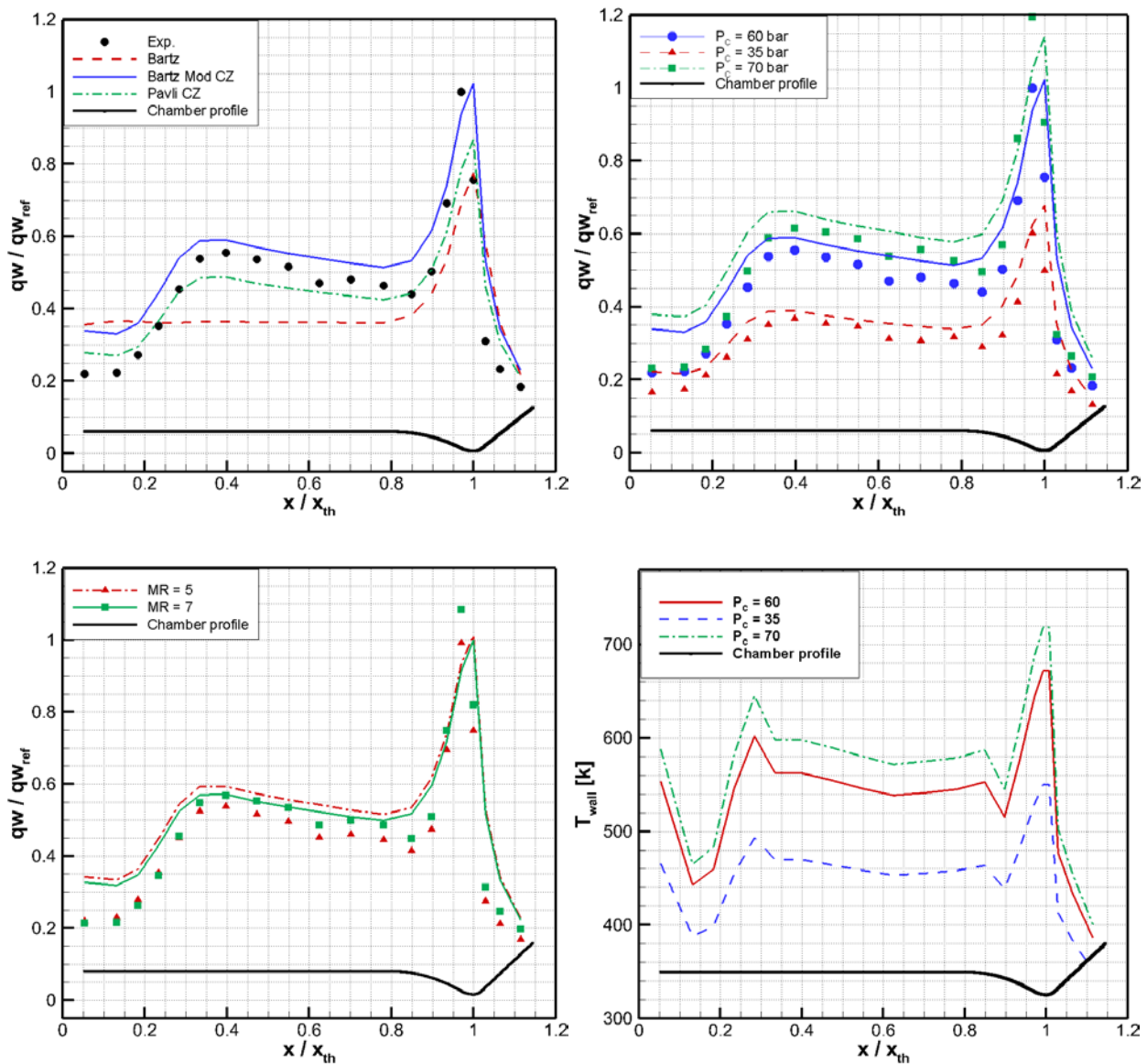


Figure 9: Heat Fluxes and wall temperatures results

The “Combustion zone” correction factor is able to represent the lower heat fluxes in the first part of the chamber. Unfortunately, the introduced correction cannot be considered predictive (that is, a correction that would give good results in a different combustion chamber and injector face): it would require experimental data with different calorimetric chambers and different injector configurations.

This is out of the scope of a 0-D/1-D investigation. The heat fluxes in the divergent part are always overpredicted. This is a characteristic of the Bartz model and needs to be kept in mind when interpreting the results. However, introducing the “Flow acceleration” correction factor is possible to achieve a better agreement with the experimental results.

Moreover, unlike the combustion zone correction factor, its behaviour is not peculiar of the experiment considered so it can be used for different chamber configurations and performance conditions. No tuning has been performed on the Bartz and Pavli parameters, the constants have been taken as $C = 0.026$, and $C = 0.023$ respectively as recommended by Bartz and Pavli.

3.2 Rocket Engine Applications

Modelling on transient phase operations of a liquid rocket engine is a challenging task because most of the engine components are required to work at extreme operating conditions; start-up and shut-down operations feature very complex phenomena such as combustion instabilities, reverse flow in the pumps, turbopumps operating far from design conditions and two-phase flows that cannot be neglected when design studies of an engine are to be performed or the assessment of the valve opening sequence has to be done. For these reasons simulations of the ignition transient phase and the shut-down phase become necessary in order to reduce the experimental tests and increase the engine safety and reliability.

Two examples of rocket engines are described hereafter: a close cycle engine, the American RL-10, and an open cycle engine, the European HM7B.

3.3 RL-10 Rocket Engine Applications [10]

3.3.1 RL-10 Engine Architecture and operation

The start-up and shut-down transient phases of the Pratt&Whitney RL-10 engine have been taken as validation case. The RL-10 engine (model RL-10A-3-3A) has been chosen for several reasons, among which the good availability of engine construction data, performance and tests results in open literature [31], [32], [33], [37]. The results of the Steady State/Design model as well as the transient (start-up and shut-down) simulations will be summarised hereafter.

The RL10A-3-3A engine is an expander cycle engine, with nominal thrust of 73.4 kN, a mixture ratio around 5, a chamber pressure of 32.7 bar and a specific impulse of 444 seconds. The fuel turbine drives both the fuel and the oxidiser pumps (the latter being driven via a gear train). The presence of a bypass valve allows the adjustment of the flow rate through the turbine and is used consequently for thrust throttling; another valve present in the oxidiser line is used for mixture ratio trimming. The RL-10 engine has been designed to offer a multiple firing capability in flight. A schematic diagram of the engine is shown in Figure 10. The RL10A-3-3A includes seven engine valves. The propellant flows to the engine can be shut off using the Fuel Inlet Shut-off Valve (FINV) and the Oxidizer Inlet Shut-off Valve (OINV). The fuel flow into the combustion chamber can be stopped by the Main Fuel Shut-off Valve (FSOV) located just upstream of the injector plenum. The fuel interstage and discharge cool-down valves (FCV-1 and FCV-2) allow overboard venting of the coolant for fuel pump cool-down during engine pre-chill. During pre-start they also provide fuel system pressure relief during engine shut-down.

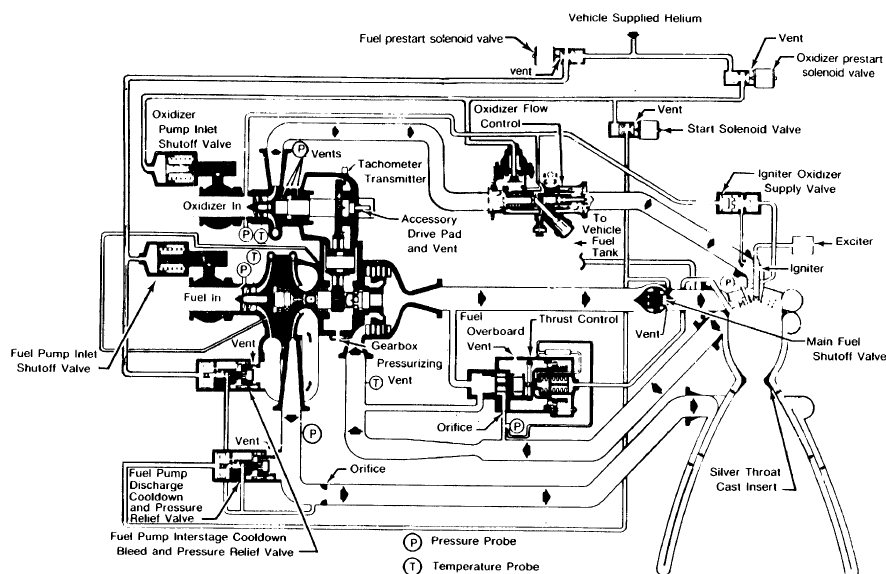


Figure 10: RL-10A-3-3A engine schematic

The Thrust Control Valve (TCV) is used to control thrust overshoot at start and maintain constant chamber pressure during steady-state operation. As combustion chamber pressure deviates from the desired value, action of the control allows the turbine bypass valve to vary the fuel flow through the turbine. The Oxidizer Control Valve (OCV) regulates the main oxidizer flow.

The Venturi upstream of the turbine is designed to help stabilize the thrust control. Ducts and manifolds in the RL10 are generally made out of stainless-steel and are not insulated [31].

3.3.2 Design/Steady State [34]

The RL-10 is an expander cycle with single turbine and geared pumps. Being a closed cycle, the cycle design is more difficult than in an open cycle, because all parameters are strongly dependent from each other. In this test case a model of the RL-10A-3-3A rocket engine system has been created and simulated using particular library derived from ESPSS library with the main purpose of simulating Rocket Engine Cycles at steady state conditions. Model results from the steady state calculations have been compared with the typical engine performance parameters at nominal operating conditions [31], [32].

Pressure drops of the main valves and junctions are taken from engine typical values [31] as well as for the tank pressure and temperature conditions. Pumps and turbine efficiency are constant and taken from open literature [31], [32]. No calibration has been adopted for the control valves: the Oxidiser Control Valve (OCV) aperture ratio has not been trimmed because the mixture ratio is assigned in the combustion chamber. Mass flows are given by combustion chamber conditions. Turbomachinery power is regulated by the Thrust Control Valve (TCV) that is open at its nominal open area ratio of 9% [33]. In Table 1 the main input data for the system is collected; in Table 2 the main system variables results are summarized, performed by the steady state model and compared with performance parameters at nominal condition.

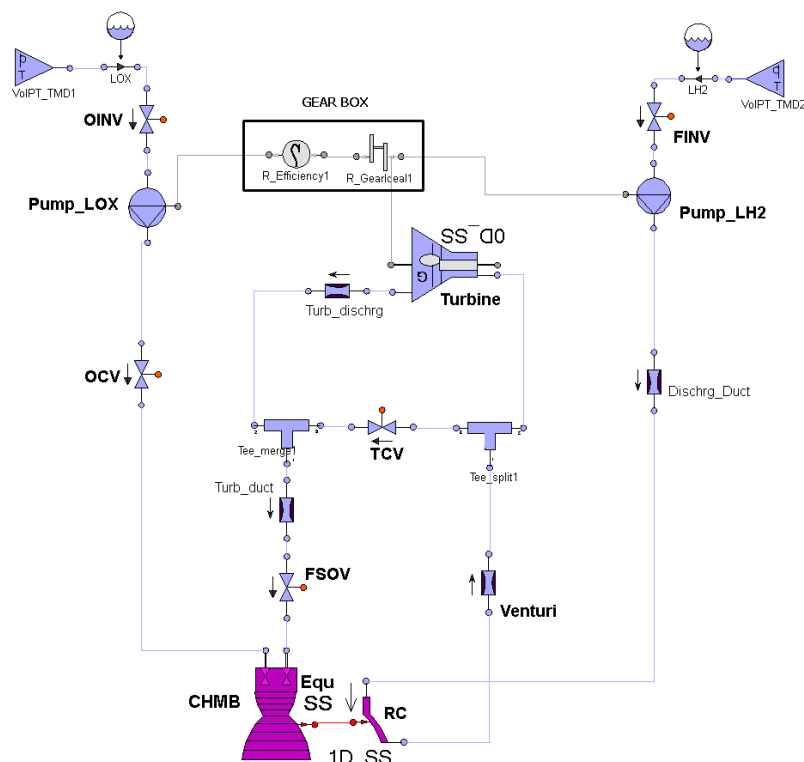


Figure 11: Schematic of the RL-10 engine using Steady State library

Table-1: RL-10A-3-3A input and initial data

Name	Description	Value	Units
$P_{in,LOX}$	Total pressure in LOX tank	2.43	bar
$T_{in,LOX}$	Total temperature in LOX tank	97.056	K
$P_{in,LH2}$	Total pressure in LH ₂ tank	1.86	bar
$T_{in,LH2}$	Total temperature in LH ₂ tank	21.44	K
P_{cc}	Nominal chamber pressure	32.75	bar
MR	Nominal chamber mixture ratio	5.055	-
n_{ch}	Numbers of cooling channels	180	-
$T_{i,cc}$	Initial chamber temperature	1000	K
m_o	Initial mass flow in the channels	2	kg/s

Table 2: RL-10A-3-3A engine system output data

Name	Nominal Value	Error
$m_{cc,ox}$ [kg/s]	13.95	0.38%
$m_{cc,fu}$ [kg/s]	2.76	0.38%
$m_{cc,t}$ [kg/s]	16.71	0.38%
m_{ch} [kg/s]	2.76	0.38%
$T_{out,ch}$ [K]	213.44	0.18%
ω_t [rpm]	31 537	0.39%
$\dot{m}_t$ [kg/s]	2.69	1.1%
W_t [kW]	588.36	3.85%
$\dot{m}_{p,LOX}$ [kg/s]	13.95	0.38%
$\omega_{p,ox}$ [rpm]	12 948	0.4%
$W_{p,ox}$ [kW]	82.026	4.07%
$\dot{m}_{p,LH2}$ [kg/s]	2.7946	0.85%
$\omega_{p,LH2}$ [rpm]	31 537	0.39%
$W_{p,LH2}$ [kW]	501.86	3.68%
$Thrust$ [kN]	73.4	2.89%
Isp [s]	444	2.5%

The chamber pressure is assigned together with the mixture ratio. The pressure cascade for both propellant lines is calculated by the model according to design parameters such as valve pressure drops. The pressure rise for the LOX pump results directly from the calculated LOX pressure cascade, whereas an algebraic loop is solved for calculating the needed H₂ pump pressure rise and the turbine pressure ratio. In parallel, several other algebraic loops are solved, determining key variables such as cooling channel outlet temperature and turbine mass flow.

The nominal fuel bypass flow ratio [33] has been fixed using the split component. The turbine evaluates the required pressure ratio and the mass flow is then evaluated via the Thrust Control Valve (TCV).

The cooling channel component evaluates iteratively with the combustion chamber component the heat fluxes and the wall temperatures. Finally, pumps power is evaluated from the required pressure rise and mass flow, and rotation speed from the given specific speed N_s .

The RL10 design model here described has proven to be very robust with respect to initial conditions. The comparison of initial conditions in Table 1 and results in Table 2 demonstrates this assertion. For example, the initial mass flow in the cooling jacket m_o is 2 kg/s, whereas the simulation result yields 2.76 kg/s. From Table 4 a very good agreement with nominal values is recognisable.

3.3.3 Transient Phases [35], [36]

The RL-10 engine starts by using the pressure difference between the fuel tank and the nozzle exit (upper atmospheric pressure), and the ambient heat stored in the metal of the cooling jacket walls. The engine "bootstraps" to full-thrust within two seconds after ignition.

A typical plot of the valve movement during engine start is shown in Figure 12. To initiate start, the FSOV is opened and the fuel-pump discharge cool-down valve (FCV-2) is closed. The interstage cool-down valve (FCV-1) remains partially open in order to avoid stalling of the fuel pump during engine acceleration.

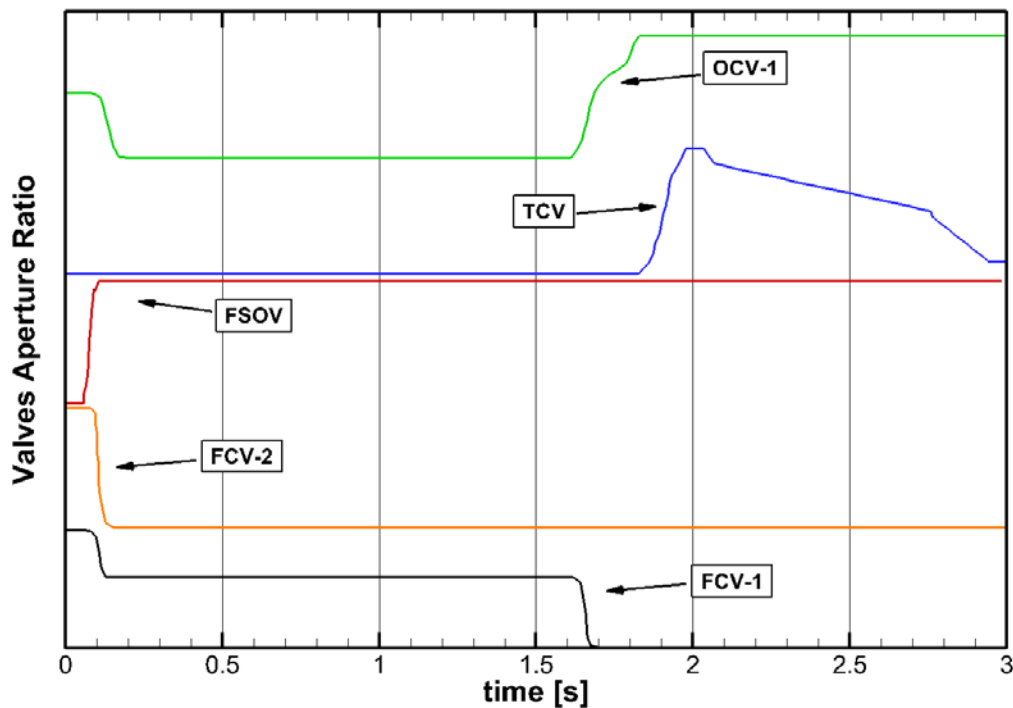


Figure 12: RL-10A-3-3A Valve schedule for Start-up Simulation

The pressure drop between the fuel inlet and the combustion chamber drives fuel through the cooling jacket, picking up heat from the warm metal. This pressure difference also drives the heated fluid through the turbine, starting rotation of the pumps, which drive more propellant into the system. At start, the OCV also closes partially, restricting the flow of oxygen into the combustion chamber in order to limit the chamber pressure and ensure a forward pressure difference across the fuel turbine after ignition of the thrust chamber.

The results of start transient simulations ("Simulation" on fig. 12 and 13) were compared with measured data of a single ground test first-burn (P2093 Run 3.02 - Test 463, "Ground Test [3]" on fig. 12 and 13) [31] and with the simulation results of a previous work ("xx_sym [3]" on fig. 12 and 13) performed by a NASA team [31].

The inlet pressure and temperature used in the model are coming from nominal operation conditions. Another important variable is the cooling jacket initial temperature, which has been set at 300 K before the pre-start phase occurs. After the pre-start simulation the cooling jacket wall temperature decreased around 240 K. It is clear that the cooling jacket wall temperatures have a great importance since they help to determine the engine start capability. The cooling jacket manifold has a lower temperature than the cooling jacket because it is partially filled by gaseous hydrogen that has not vented overboard via the fuel discharge

valve (FCV-2).

In the simulation the ignition occurs when the propellant mixture ratio inside the combustion chamber reaches a value lower than 30 (as shown in Ref.[37]), at around 0.3 seconds as in the ground test. Figure 13(a) shows the comparison between measured and predicted chamber pressure. The model matches the measured time-to-accelerate to within approximately 92 milliseconds (the "time-to-accelerate" is defined here as the time from 0 seconds up to the time at which the chamber pressure reaches 13.79 bar (200 psia)). The difference with the previous model of the RL-10A-3-3A transient start-up [31] is remarkable.

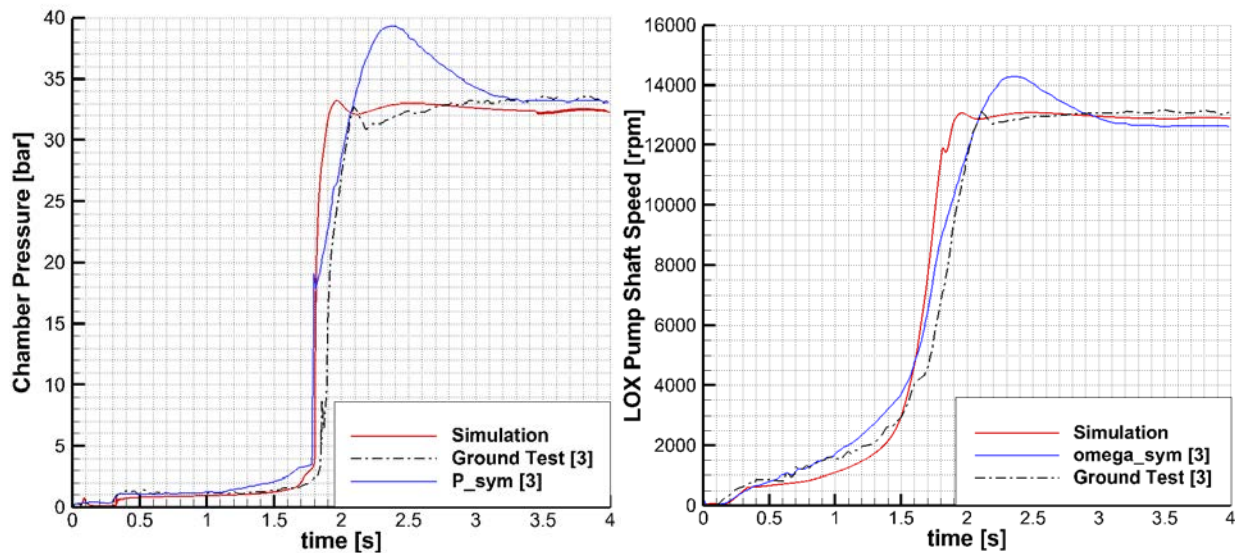


Figure 13: (a) Chamber Pressure; (b) LOX Pump Shaft Speed

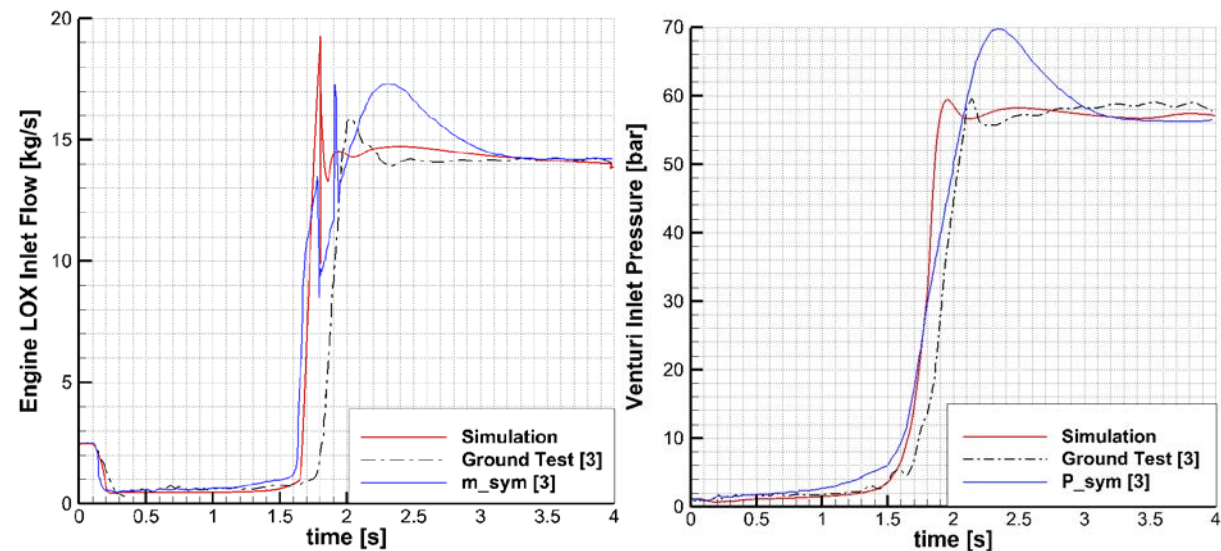


Figure 14: (a) LOX Pump Discharge Pressure; (b) Venturi inlet Pressure

In Figure 13 (b) the LOX pump rotational speed is shown: the simulation result is in good agreement with the experimental result. The difference in the rate of change of the pump speed between the simulation and experiment may be due to the uncertainty in the pump inertia distribution.

Figures 14 (a) depicts the LOX pump outlet pressure evolution. The simulation exhibits some sharp transients before reaching steady-state conditions; these seem to be due to fluid compressions and phase changes that occur when the OCV suddenly opens. These transients are steeper than the measured data

probably because the dynamic behaviour of the OCV valve plays an important role in the fluid dynamics during pressure rise.

The last measured point in the engine was the pressure at the Venturi inlet. This is illustrated in Figure 14 (b): it is evident that the simulation evolution is in a very good agreement with experimental data.

The RL-10 engine switches off at the end of its mission, after the steady state phase. The Fuel Shut-off Valve (FSOV) and the Fuel Inlet Valve (FINV) close as the FCV-1 and FCV-2 valves open, allowing fuel to drain out of the system through the overboard vents. The combustion process is soon starved of fuel and the flame extinguishes. The Oxidiser Control Valve (OCV) and the Oxidiser Inlet valve (OINV) begin to close next, cutting off the flow of oxygen through the engine. The turbopump decelerates due to friction losses and drag torque created by the pumps as they evacuate the remaining propellants from the system. A typical plot of the valve movement during engine shut-down is shown in Figure 15.

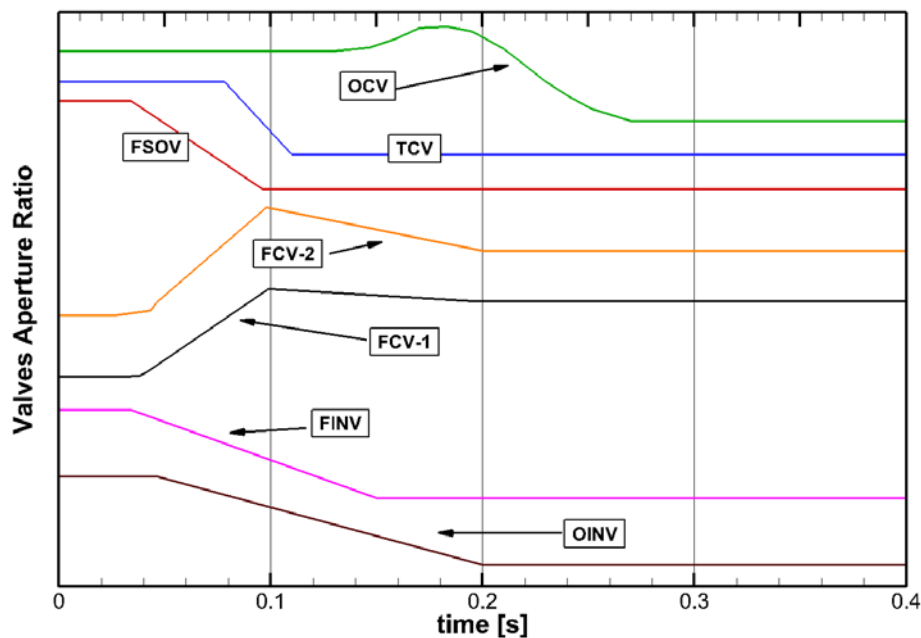


Figure 15: RL-10A-3-3A Valve schedule for Shut-down Simulation

During the engine shut-down, a different combination of off-design conditions appears to exist, including pump cavitation and reverse flow. Proper simulation of these effects is complicated by their interaction with each other. The cavitation causes the pump performance to degrade rapidly until the pump cannot prevent the reverse flow of fluid as it comes backward through the cooling jacket. When the reversed flow reaches the closed fuel inlet valve, however, extreme transients of pressure and flow are created. Similar effects are encountered in the LOX pump during shut-down as well.

The pump head and torque performance characteristics during, this period of operation are, of course, not extensively documented in test data. The generic pump characteristics found in Reference [38] have been used to extend the performance maps for cavitation and reverse flow. Although the engine start-up and shut-down models use the same pump performance maps (which should be able to cover all the pump regimes), the cavitation and reverse flow effects also require additional modelling effort that has not been implemented into this model yet.

The results of shut-down simulations (“Simulation” on the plots) were compared with measured data of a single ground test (P2093 Run 8.01 - Test 468, “Ground Test [3]” on the plots) [31] and with the simulation results of a previous work (“xx_sym [3]” on the plots) performed by the NASA team [31].

Figure 16 (a) illustrates the combustion chamber pressure trend. Once the FSOV starts to close the chamber pressure decreases and this happens in both the simulations and the experimental data, showing a good

agreement between them. Figure 16 (b) shows predicted and measured pump speed for the Oxidiser propellant side. The discrepancies from the two models and the ground test measured data are imputable to uncertainties on the inlet conditions and initial operating point as well as on a precise distribution of the turbopump assemblies' inertia.

Figures 17 (a) depicts the LOX pump outlet pressure. Once the FSOV valve starts to close, the outlet pump pressure decreases because of the minor power delivered by the turbine.

The oxygen mass flow behaviour (Figure 17 (b)) is mainly function of the pump behaviour; from analyses performed varying the opening/closing time of the valves, the role of the FCV valves becomes much more evident. The opening of the FCV valves decreases the turbine power, hence decreasing the propellant mass flow rate in the system in order to avoid mass flow rate surges of oxygen at the FSOV closure. In the end, the complete shut-off of the OINV valve extinguishes the propellant flow rate.

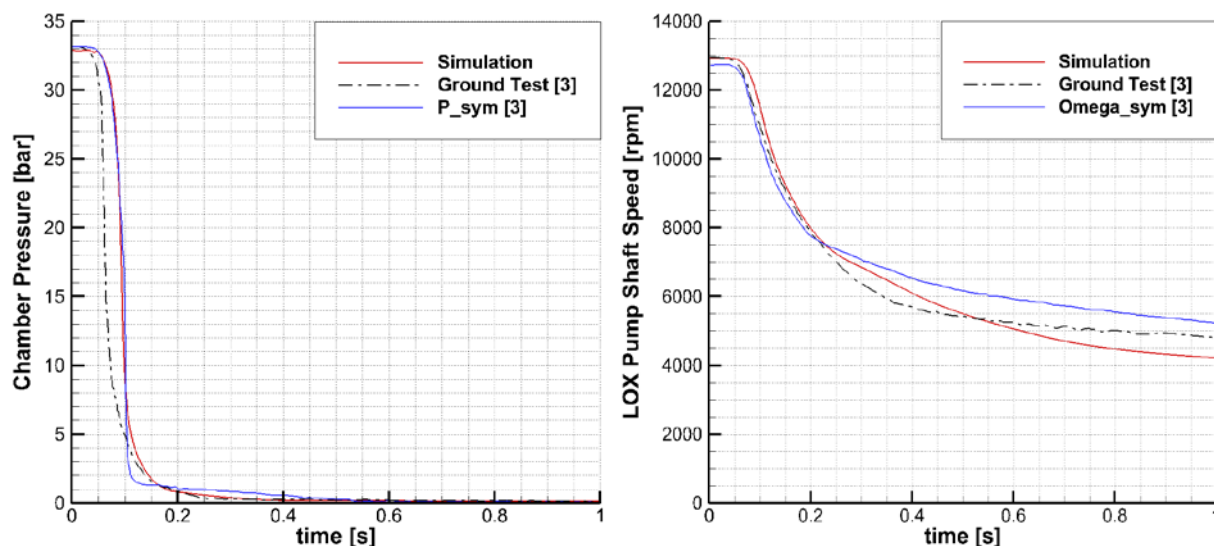


Figure 16: (a) Chamber Pressure; (b) LOX Pump Shaft Speed

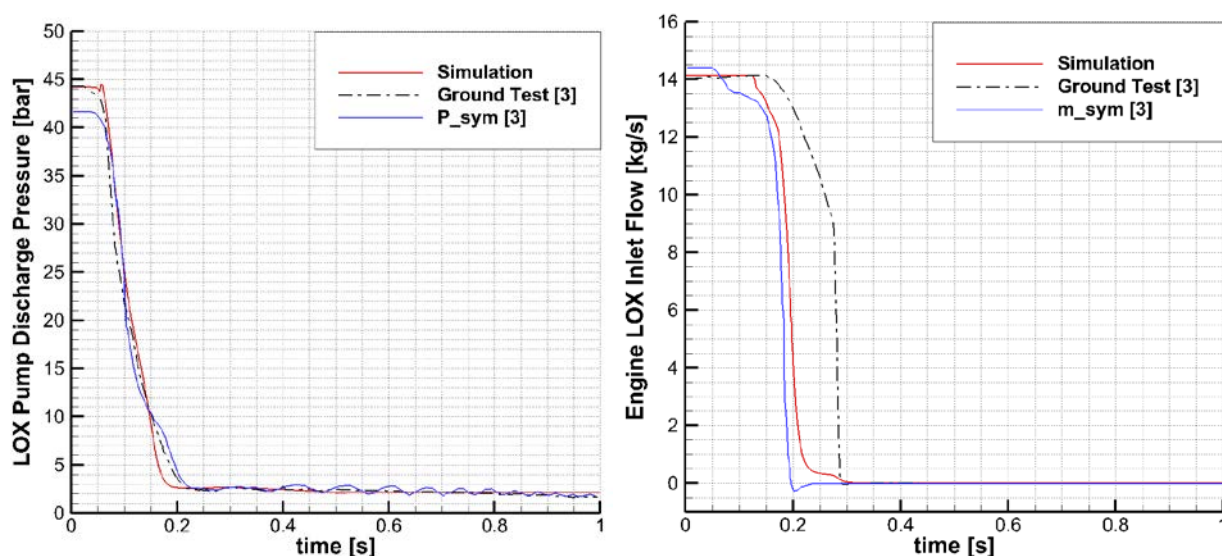


Figure 17: (a) LOX Pump Discharge Pressure; (b) LOX Engine inlet mass flow

3.4 HM7B Rocket Engine Applications

3.4.1 HM7B Engine Architecture and operation

The bi-propellant and cryogenic upper stage ESC-A is the latest improvement of the ARIANE 5 and can be used instead of the EPS in case re-ignition is not needed. The ESC-A is propelled by the HM7B motor which was already developed for ARIANE 1 3rd stage and successfully flown on ARIANE 1 to 4 launchers. A simplified sketch is shown below:

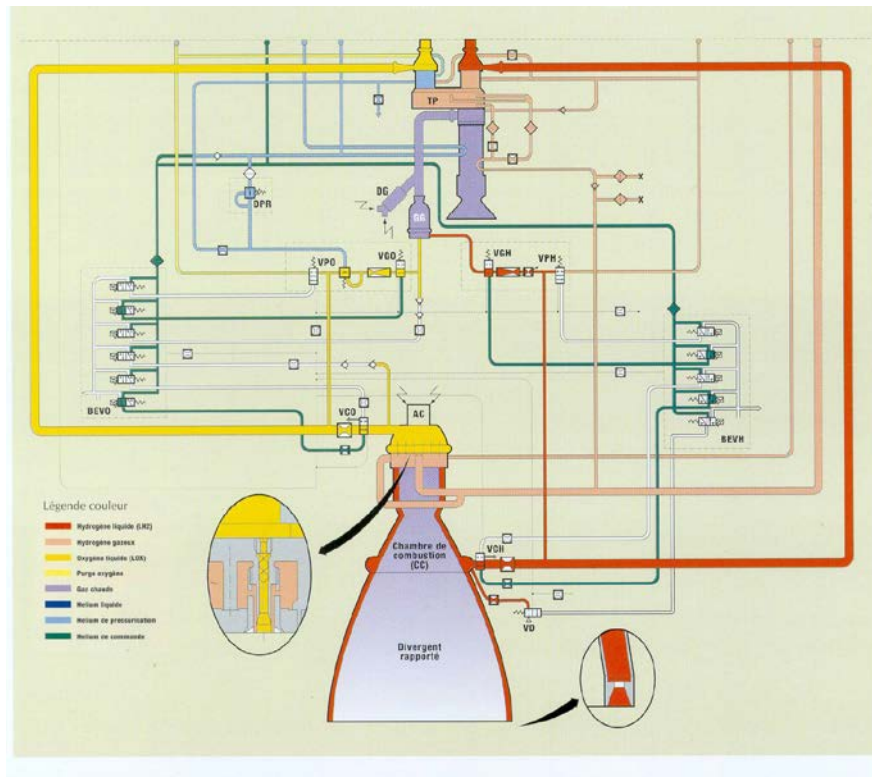


Figure 18: HM7B engine schematic [40]

The HM7B is a gas generator engine with single turbine and geared pumps. The engine is composed of the following main components:

- 2 propellant feed lines incl. tank outlet valves
- 1 gas generator
- 2 pyro igniters
- 2-stage turbine with heater incl. expansion nozzle
- 2 single stage radial pumps
- 1 gear box incl. ventilation system
- lubricant system
- 2 gas generator injector blocks
- 6 engine valves
- 1 regenerative cooled combustion chamber incl. gaseous hydrogen tap-off
- 1 dump-cooled supersonic nozzle extension
- entire piping system

The HM7B engine has nominal thrust of 64.8 kN, a mixture ratio around 5, a chamber pressure of 36.6 bar and a specific impulse of 445.6 seconds. The fuel turbine drives both the fuel and the oxidiser pumps (the latter being driven via a gear train).

The HM7B includes six engine valves. The fuel flow into the combustion chamber can be stopped by the VCH located just upstream of the cooling jacket. The VCO regulates the main oxidizer flow, while the VGO and the VGH valves regulate respectively the oxygen and the hydrogen flows towards the gas generator.

3.3.2 Design/Steady State [34]

The HM7B rocket engine system has been created and simulated using a particular library derived from ESPSS library with the main purpose of simulating Rocket Engine Cycles at steady state conditions. Model results from the steady state calculations have been compared with the typical engine performance parameters at nominal operating conditions.

Pressure drops of the main valves and junctions are taken from engine typical values as well as for the tank pressure and temperature conditions. Pumps and turbine efficiency are kept constant. A schematic of the HM7B engine created with the Steady State library is shown in Figure 19.

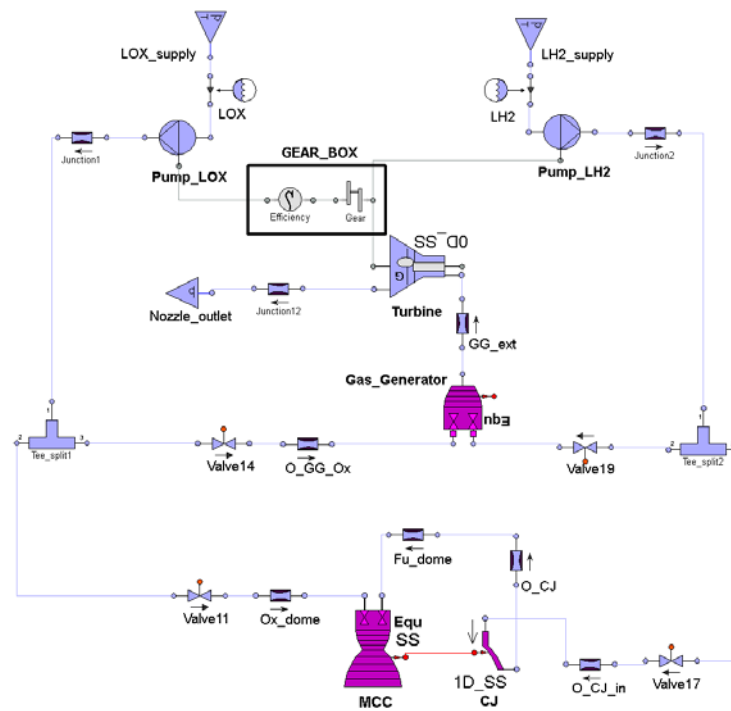


Figure 19: HM7B engine system schematic

In Table 3 the main input data for the system is collected; In Table 4 the main system variables results are summarized and compared with nominal data. Where nominal values are shown, they are taken from the open literature [40].

Pressure drop has been fixed in each valve and junction as well as turbomachinery efficiency and gas generator mixture ratio. For the turbine, the pressure ratio has been given, while the pump specific speed N_s has been calculated from design data.

The main chamber pressure and mixture ratio are fixed. Propellant mass flows to the main chamber are calculated, and fed back to the upstream components. The gas generator mass flow is not fixed. Only its mixture ratio is fixed, in accordance with the maximum allowable temperature in the turbine. The mass flow is then calculated by an algebraic equation system resulting automatically from the connection of gas generator and turbopumps. The needed shaft power drives the total gas generator mass flow rate, since the turbine pressure ratio is given by design. Initial values such as mass flow rates and shaft speed are chosen by rough engineering assessments.

Table 3: HM7B input and initial data

Name	Description	Value	Units
$P_{in,LOX}$	Total pressure in LOX tank	2.0	bar
$T_{in,LOX}$	Total temperature in LOX tank	91.2	K
$P_{in,LH2}$	Total pressure in LH ₂ tank	3.0	bar
$T_{in,LH2}$	Total temperature in LH ₂ tank	21.0	K
P_{cc}	Nominal chamber pressure	36.6	bar
MR	Nominal chamber mixture ratio	5.	-
n_{ch}	Numbers of channels	128	-
$T_{i,cc}$	Initial chamber temperature	1000	K
P_o	Initial total pressure in the channels	50	bar
T_o	Initial total temperature in the channels	30	K
m_o	Initial mass flow in the channels	2	kg/s
$P_{c,gg}$	Initial Gas Generator pressure	20	bar
T_{ch}	Initial Gas Generator temperature	900	K
rpm _{ox}	Initial LOX pump speed	1000	rpm
$N_{s,ox}$	LOX pump specific speed	10.95	-
rpm _{fu}	Initial H ₂ pump speed	6000	rpm
$N_{s,fu}$	H ₂ pump specific speed	9.29	-
$m_{o,tu}$	Initial Turbine mass flow	0.2	kg/s

Table 4: HM7B engine system output data ¹

Name	Nominal Value ¹	Error
$\dot{m}_{ox,gg}$ [kg/s]		0.59%
$\dot{m}_{fu,gg}$ [kg/s]		0.23%
$\dot{m}_{t,gg}$ [kg/s]		0.3%
P_{gg} [bar]		0%
T_{gg} [K]		1.03%
$m_{cc,ox}$ [kg/s]		0.88%
$m_{cc,fu}$ [kg/s]		0.83%
$m_{cc,t}$ [kg/s]	14.86	0.87%
m_{ch} [kg/s]		2.36%
ΔP_{ch} [bar]		0.5%
$T_{out,ch}$ [K]		6.7%
ω_t [rpm]	60500	0.16%
τ_t [N·m]		4.48%
W_t [W]		4.32%
$T_{in,t}$ [K]		11.3%
$\dot{m}_{p,LOX}$ [kg/s]		0.96%
ΔP_{LOX} [bar]	48	0.41%
$\omega_{p,ox}$ [rpm]	13000	0.16%
$\tau_{p,ox}$ [N·m]		1.39%
$\dot{m}_{p,LH2}$ [kg/s]		7.56%
ΔP_{LH2} [bar]	52	1.29%
$\omega_{p,fu}$ [rpm]	60500	0.16%
$\tau_{p,fu}$ [N·m]		6.0%

From Table 4 a very good agreement with nominal data is recognisable. Few parameters have a higher percentage error: the fuel mass flow rate in the pump does not take into account the dump and the tap-off mass flow rate vented from the engine. The cooling jacket exit temperature takes into account the temperature increase in the injector dome. If compared to the LH₂ injector dome temperature, the error

¹ see Ref. [40]

decreases to 2.47%. Turbomachinery parameters show quite good results; the differences are mainly due to the turbine inlet temperature that is lower than the nominal one.

3.4.2 Transient Phases [39]

This test case was used during the ESPSS Industrial Evaluation from Astrium Bremen to validate the ESPSS library for liquid rocket engine cycles [41]. This means that most of the data are not available for publishing. Figure 20, shows the ESPSS schematic according to the HM7B motor.

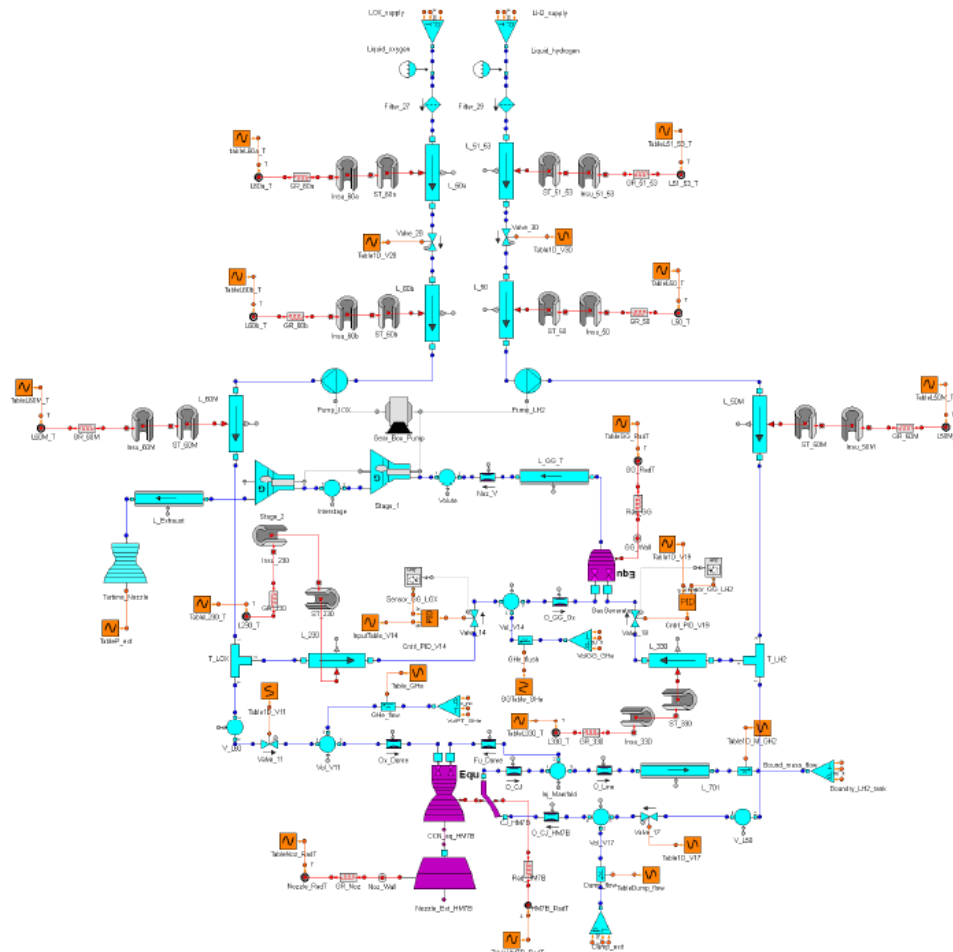


Figure 20: ESPSS Schematic [39] Representation of HM7B Engine

Simplifications are introduced by not considering the pressurization gas circuit for the pilot valves. The main engine valves and the propellant feed line valves are controlled by a time-dependent data table. The gas generator injection blocks for LOX and LH2 are simplified to PID controlled valves with a regulated flow rate each introduced as a set value. The Helium heating circuit is also not considered as well as the gaseous Helium and Hydrogen flushing of the gear box and its lubrication. The dump cooling flow of the supersonic nozzle extension is only considered as wasted flow by a respective adjusted mass flow rate. With these simplifications the HM7B schematic was built.

The validation covers the engine system start-up transient. With the discussed set of parameters the engine model is now capable to obtain the steady state values. Unknown values e.g. pressure losses in the lines are

thereby easy to determine. A comparison is made to transient flight measurement transducers of launch A5 L521;

Figure 21 (a) shows the thrust chamber pressure evolution during start-up inside the thrust chamber. In the simulation the pressure increase starts slightly later compared to the measurement, which seems to be a result of the chosen LOX chamber valve time constant of 0.01s. The pressure increase follows well the measurement. The still existing slight differences can be further reduced by on-going parameter adjustment. The gas generator chamber pressure is during pyro-starter operation in good equivalence to the respective measurement, see Figure 21 (b). The slight decrease of starter pressure evolution is related to decrease in mass flow rate and could be adapted by defining a time-dependent mass flow rate in the model. A slight over-oscillation can be still remarked.

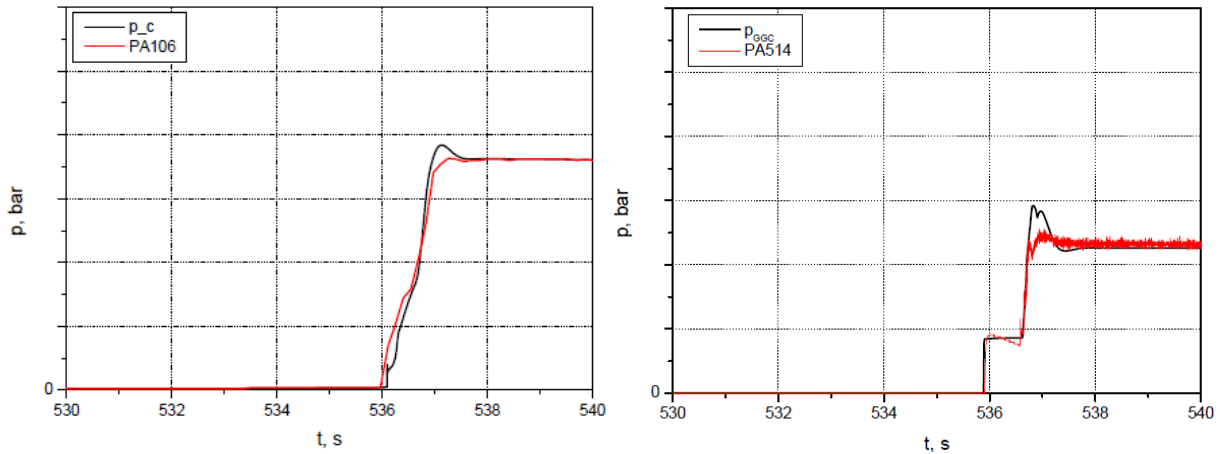


Figure 21: (a) Thrust Chamber Pressure P_c ; (b) Gas Generator Chamber Pressure P_{GGC}

This "over-oscillation" is clearly an effect of the pumps outlet pressures see 22 (a) and 22 (b). When changing from pyro-starter operation mode to LH2/LOX combustion mode, it can be remarked that the simulated pressure curves increase faster.

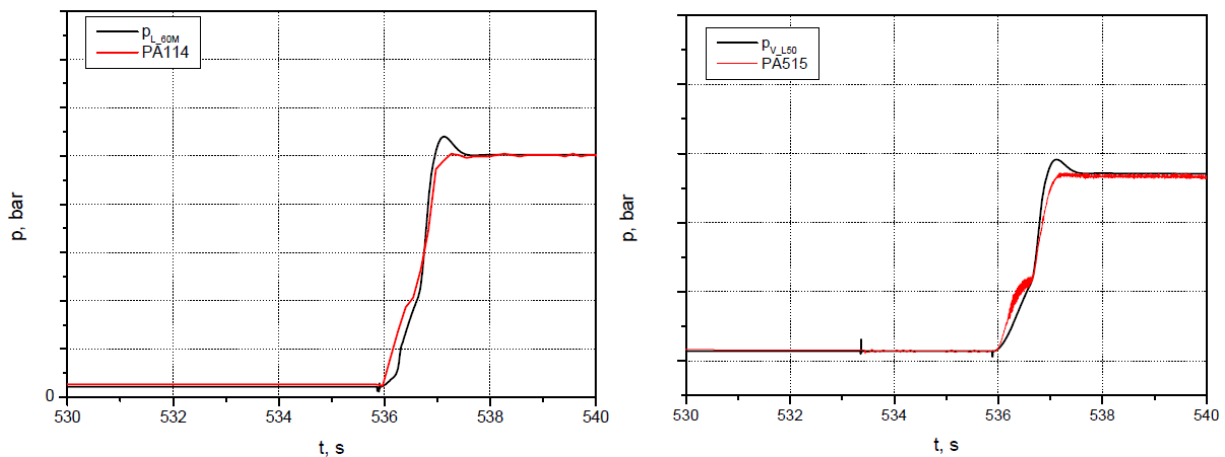


Figure 22: (a) Outlet LOX Pump Pressure P_{L60M} ; (b) Outlet LH2 Pump Pressure P_{V50}

CONCLUSIONS

Transient phenomena in liquid rocket engines, ranging from combustion high frequency instabilities to water hammer effects in the feed lines, and which potentially result in system failures, drive the necessity to dedicate special attention to transient phases.

The simulation of the complex flow behaviour in engine components and components assemblies is therefore required. The development of models capable of simulating in a more accurate way respect to previous models liquid rocket engine components and propulsion system resulted in the work performed to create the ESPSS libraries. They can simulate the major liquid rocket engine components: pipes, valves, injector domes, injectors, turbopumps, combustion devices and nozzles.

The power of this tool has been proved here presenting different simulations, ranging from academic test case (the shock –tube test case), through components level performance evaluation (hot gas side heat transfer in combustion chambers) till system level simulations showing transient phases (start-up and shut-down) of entire liquid rocket engine systems (RL-10 and HM7B engines).

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